UNITED STATES INTERNATIONAL UNIVERSITY

AFRICA

DSA 3050A: DAX Lab Assignment

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Dataset Source Link

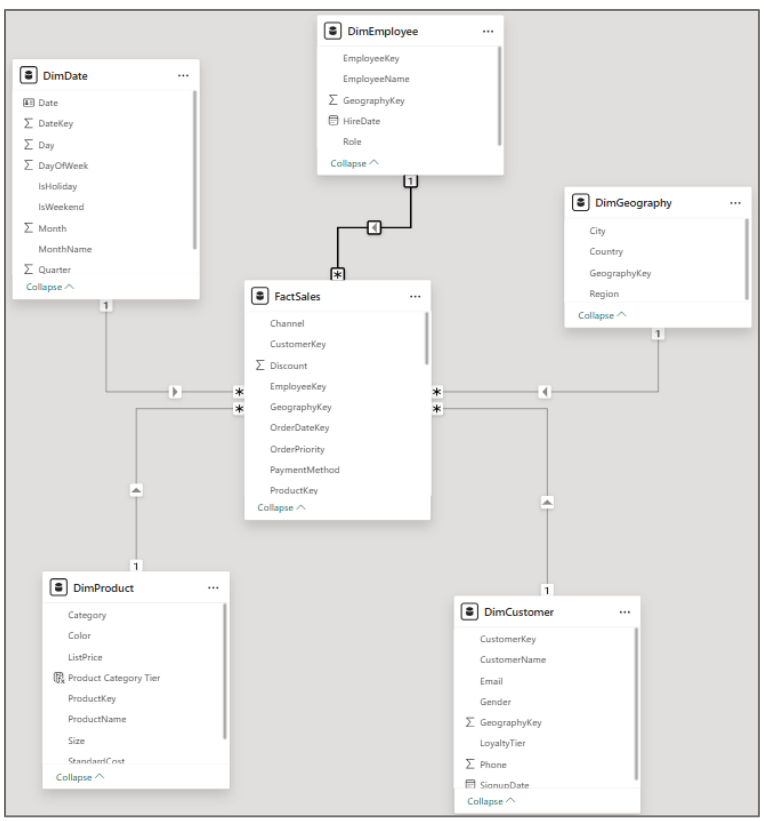
[Kaggle - Complete DAX Practice Dataset]

SECTION A: DATA MODEL VALIDATION

1. Dataset Description

The dataset used for this assignment is the Complete DAX Practice Dataset, a synthetic e-commerce dataset simulating a multi-year sales environment (2018-2024). It consists of approximately 100,000 rows in the fact table, making it well-suited for high-volume data analysis. The data is organized in a Star Schema with a central FactSales table and five dimension tables (DimProduct, DimCustomer, DimDate, DimEmployee, and DimGeography), allowing for granular slicing across categories, geography, and time.

2. Model View:

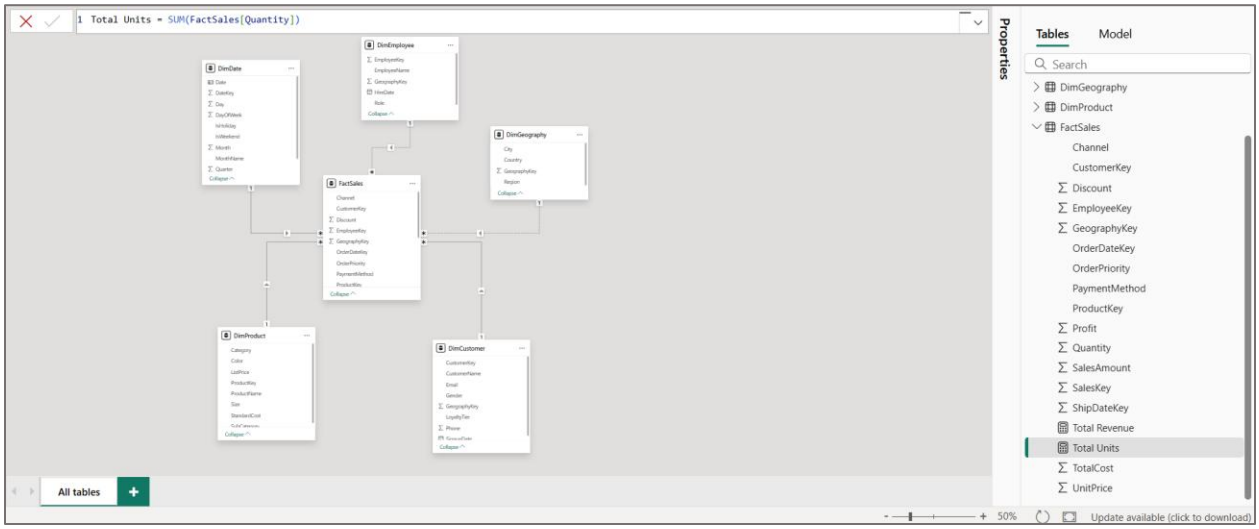


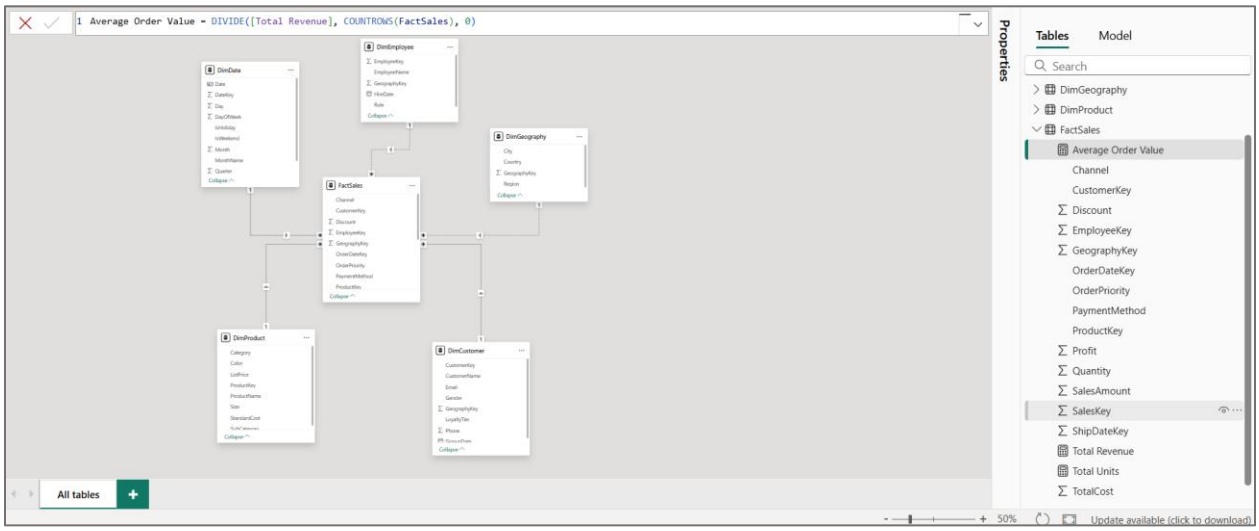
3. One Challenge Encountered

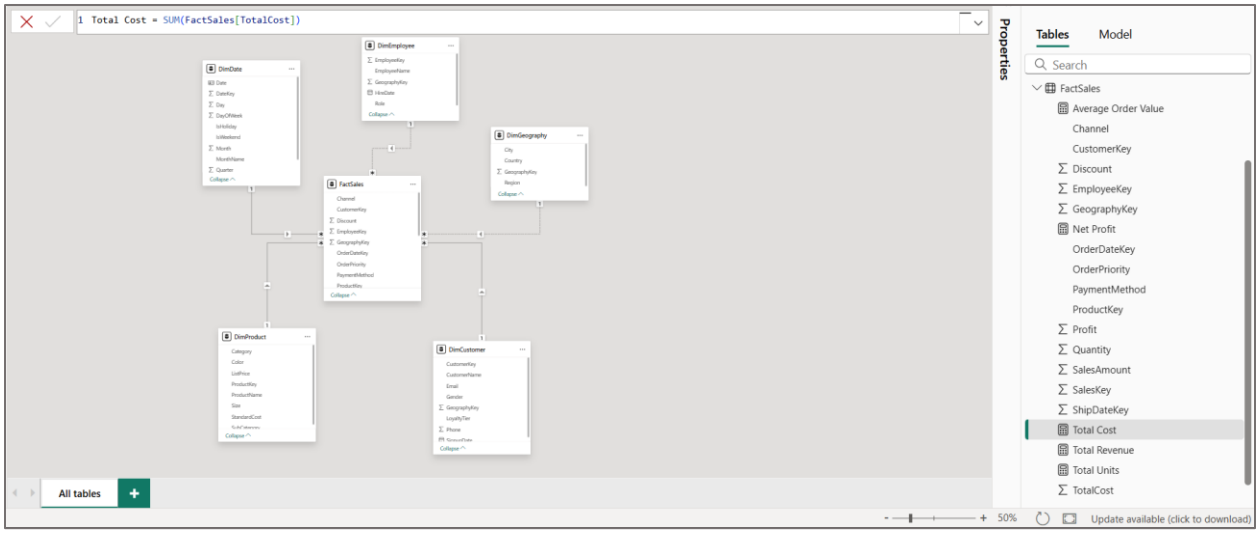
One significant challenge was managing the Filter Context for regional comparisons. Initially, the 'Total Revenue' measure was showing global totals across all rows. I resolved this by identifying a broken relationship in the Model View and updating the Cross Filter Direction to 'Single' (from Dimension to Fact), ensuring that filters propagate correctly from the Geography table to the Sales table.

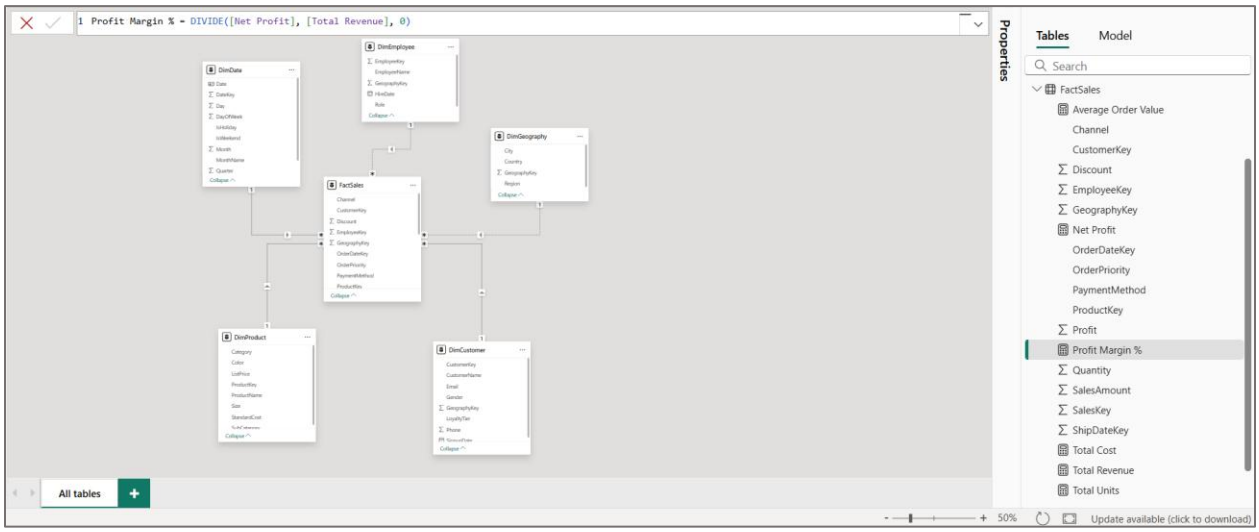
SECTION B: CORE DAX MEASURES

This section establishes the foundational metrics of the data model. Using standard aggregation functions like SUM and COUNT, I created base measures for Total Revenue, Total Cost, and Total Transactions. These measures form the building blocks for the more complex calculations used in later sections, ensuring a 'single version of truth' for all financial reporting.







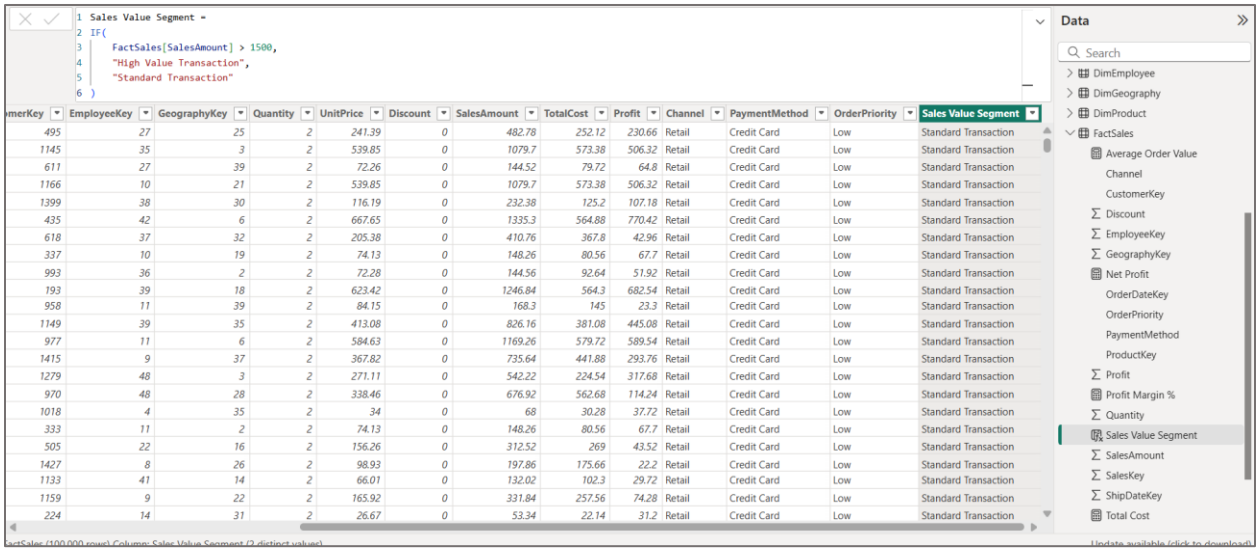


Responding correctly to slicers

Category	Total Revenue	Total Cost	Total Units	Average Order Value	Net Profit	Profit Margin %
Accessories	1,865,039.71	1,134,906.02	6722	824.87	730,133.69	0.39
Clothing	2,069,535.94	1,347,914.23	7210	846.78	721,621.71	0.35
Electronics	1,497,121.63	913,347.81	6114	728.17	583,773.82	0.39
Furniture	1,378,789.51	914,410.72	6250	665.12	464,378.79	0.34
Kitchen	1,883,941.21	1,274,995.76	9149	619.92	608,945.45	0.32
Outdoors	1,401,757.12	900,649.32	7062	590.96	501,107.80	0.36
Total	**10,096,185.12**	**6,486,223.86**	**42507**	**708.75**	**3,609,961.26**	**0.36**

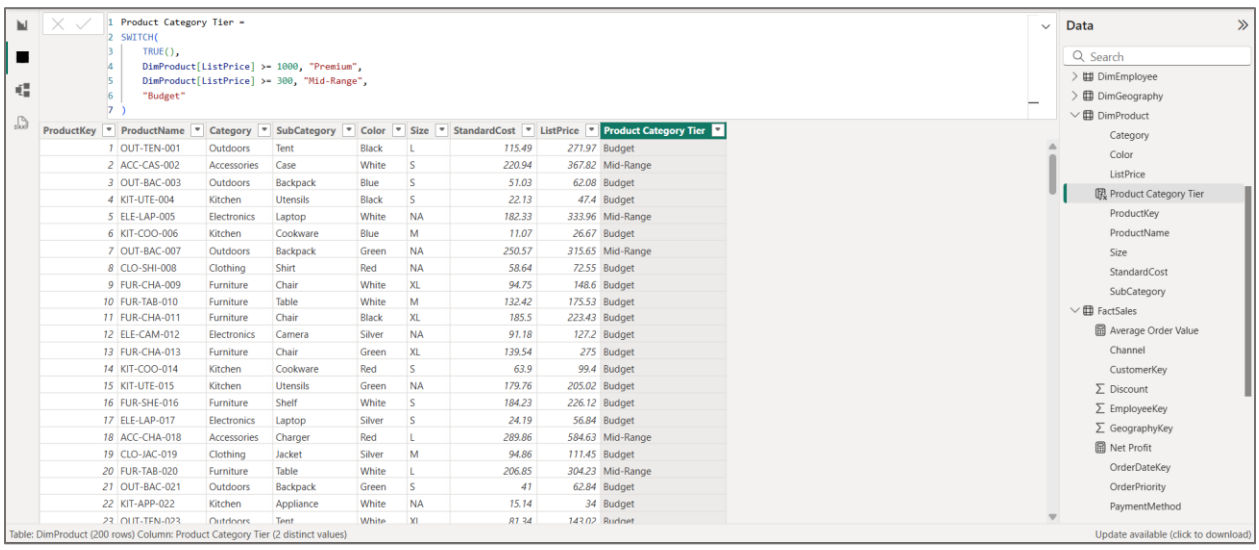
Year

2020



```
Sales Value Segment =
IF(
    FactSales[SalesAmount] > 1500,
    "High Value Transaction",
    "Standard Transaction"
)
```

omerKey	EmployeeKey	GeographyKey	Quantity	UnitPrice	Discount	SalesAmount	TotalCost	Profit	Channel	PaymentMethod	OrderPriority	Sales Value Segment
495	27	25	2	241.39	0	482.78	252.12	230.66	Retail	Credit Card	Low	Standard Transaction
1145	35	3	2	539.85	0	1079.7	573.38	506.32	Retail	Credit Card	Low	Standard Transaction
611	27	39	2	72.26	0	144.52	79.72	64.8	Retail	Credit Card	Low	Standard Transaction
1166	10	21	2	539.85	0	1079.7	573.38	506.32	Retail	Credit Card	Low	Standard Transaction
1399	38	30	2	116.19	0	232.38	125.2	107.18	Retail	Credit Card	Low	Standard Transaction
435	42	6	2	667.65	0	1335.3	564.88	770.42	Retail	Credit Card	Low	Standard Transaction
618	37	32	2	205.38	0	410.76	367.8	42.96	Retail	Credit Card	Low	Standard Transaction
337	10	19	2	74.13	0	148.26	80.56	67.7	Retail	Credit Card	Low	Standard Transaction
993	36	2	2	72.28	0	144.56	92.64	51.92	Retail	Credit Card	Low	Standard Transaction
193	39	18	2	623.42	0	1246.84	564.3	682.54	Retail	Credit Card	Low	Standard Transaction
958	11	39	2	84.15	0	168.3	145	23.3	Retail	Credit Card	Low	Standard Transaction
1149	39	35	2	413.08	0	826.16	381.08	445.08	Retail	Credit Card	Low	Standard Transaction
977	11	6	2	584.63	0	1169.26	579.72	589.54	Retail	Credit Card	Low	Standard Transaction
1415	9	37	2	367.82	0	735.64	441.88	293.76	Retail	Credit Card	Low	Standard Transaction
1279	48	3	2	271.11	0	542.22	224.54	317.68	Retail	Credit Card	Low	Standard Transaction
970	48	28	2	338.46	0	676.92	562.68	114.24	Retail	Credit Card	Low	Standard Transaction
1018	4	35	2	34	0	68	30.28	37.72	Retail	Credit Card	Low	Standard Transaction
333	11	2	2	74.13	0	148.26	80.56	67.7	Retail	Credit Card	Low	Standard Transaction
505	22	16	2	156.26	0	312.52	269	43.52	Retail	Credit Card	Low	Standard Transaction
1427	8	26	2	98.93	0	197.86	175.66	22.2	Retail	Credit Card	Low	Standard Transaction
1133	41	14	2	66.01	0	132.02	102.3	29.72	Retail	Credit Card	Low	Standard Transaction
1159	9	22	2	165.92	0	331.84	257.56	74.28	Retail	Credit Card	Low	Standard Transaction
224	14	31	2	26.67	0	53.34	22.14	31.2	Retail	Credit Card	Low	Standard Transaction



```
Product Category Tier =
SWITCH(
    TRUE(),
    DimProduct[ListPrice] >= 1000, "Premium",
    DimProduct[ListPrice] >= 300, "Mid-Range",
    "Budget"
)
```

ProductKey	ProductName	Category	SubCategory	Color	Size	StandardCost	ListPrice	Product Category Tier
1	OUT-TEN-001	Outdoors	Tent	Black	L	115.49	271.97	Budget
2	ACC-CAS-002	Accessories	Case	White	S	220.94	367.82	Mid-Range
3	OUT-BAC-003	Outdoors	Backpack	Blue	S	51.03	62.08	Budget
4	KIT-UTE-004	Kitchen	Utensils	Black	S	22.13	47.4	Budget
5	ELE-LAP-005	Electronics	Laptop	White	NA	182.33	333.96	Mid-Range
6	KIT-COO-006	Kitchen	Cookware	Blue	M	11.07	26.67	Budget
7	OUT-BAC-007	Outdoors	Backpack	Green	NA	250.57	315.65	Mid-Range
8	CLO-SHI-008	Clothing	Shirt	Red	NA	58.64	72.55	Budget
9	FUR-CHA-009	Furniture	Chair	White	XL	94.75	148.6	Budget
10	FUR-TAB-010	Furniture	Table	White	M	132.42	175.53	Budget
11	FUR-CHA-011	Furniture	Chair	Black	XL	185.5	223.43	Budget
12	ELE-CAM-012	Electronics	Camera	Silver	NA	91.18	127.2	Budget
13	FUR-CHA-013	Furniture	Chair	Green	XL	139.54	275	Budget
14	KIT-COO-014	Kitchen	Cookware	Red	S	63.9	99.4	Budget
15	KIT-UTE-015	Kitchen	Utensils	Green	NA	179.76	205.02	Budget
16	FUR-SHE-016	Furniture	Shelf	White	S	184.23	226.12	Budget
17	ELE-LAP-017	Electronics	Laptop	Silver	S	24.19	56.84	Budget
18	ACC-CHA-018	Accessories	Charger	Red	L	289.86	584.63	Mid-Range
19	CLO-JAC-019	Clothing	Jacket	Silver	M	94.86	111.45	Budget
20	FUR-TAB-020	Furniture	Table	White	L	206.85	304.23	Mid-Range
21	OUT-BAC-021	Outdoors	Backpack	Green	S	41	62.84	Budget
22	KIT-APP-022	Kitchen	Appliance	White	NA	15.14	34	Budget
23	OUT-TEN-023	Outdoors	Tent	White	XL	81.34	143.02	Budget

Table: DimProduct (200 rows) Column: Product Category Tier (2 distinct values)

SECTION C: CALCULATED COLUMNS

Correct Row-Level Logic

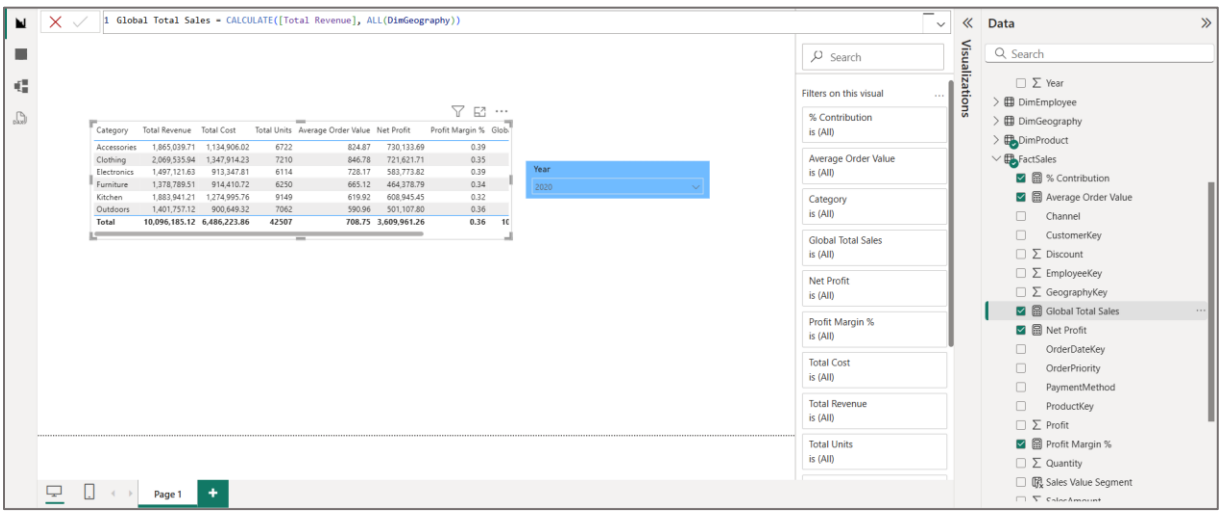
The row-level logic was implemented using the SWITCH(TRUE(), ...) and IF functions. Unlike measures, these calculations are performed for every individual record in the dataset during data load. For example, in the Product Category Tier column, the engine evaluates the ListPrice of each specific row against the defined conditions to assign a static category. This ensures that the categorization is 'baked into' the data model, allowing for high-performance filtering in the final dashboard.

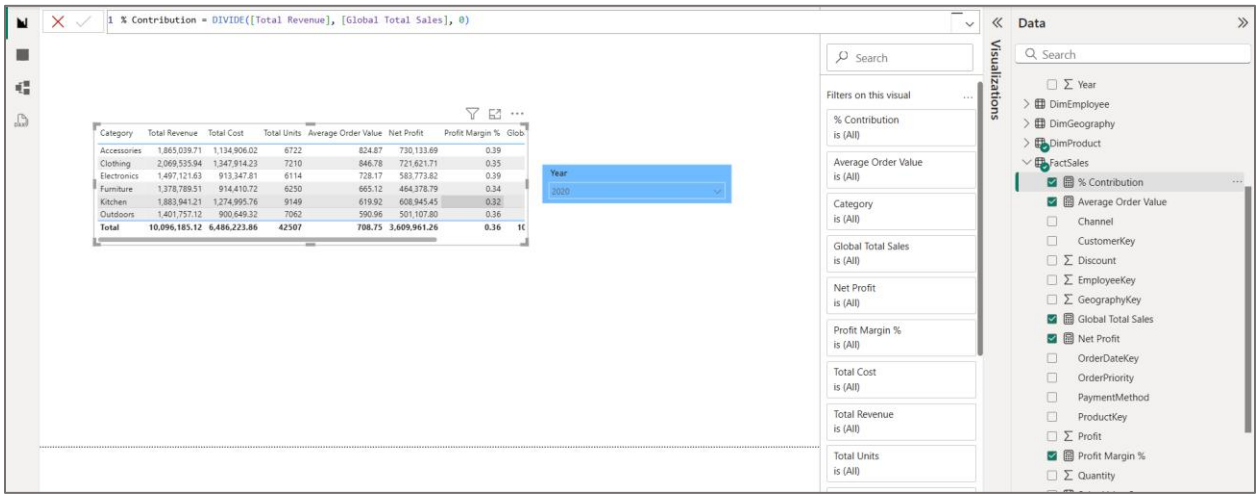
Justified Business Thresholds

The thresholds for classification were selected based on an analysis of the price distribution within the electronics dataset:

- Premium (>= $1000): This threshold captures high-ticket items like high-end laptops and professional cameras, which represent the luxury segment of the inventory.
- Mid-Range (>= $300): This captures standard consumer electronics like tablets and mid-tier smartphones, which form the bulk of the revenue.
- Budget (< $300): This covers accessories and entry-level gadgets.

By setting these specific thresholds, the business can distinguish between high-volume/low-margin sales and low-volume/high-margin 'Premium' sales.

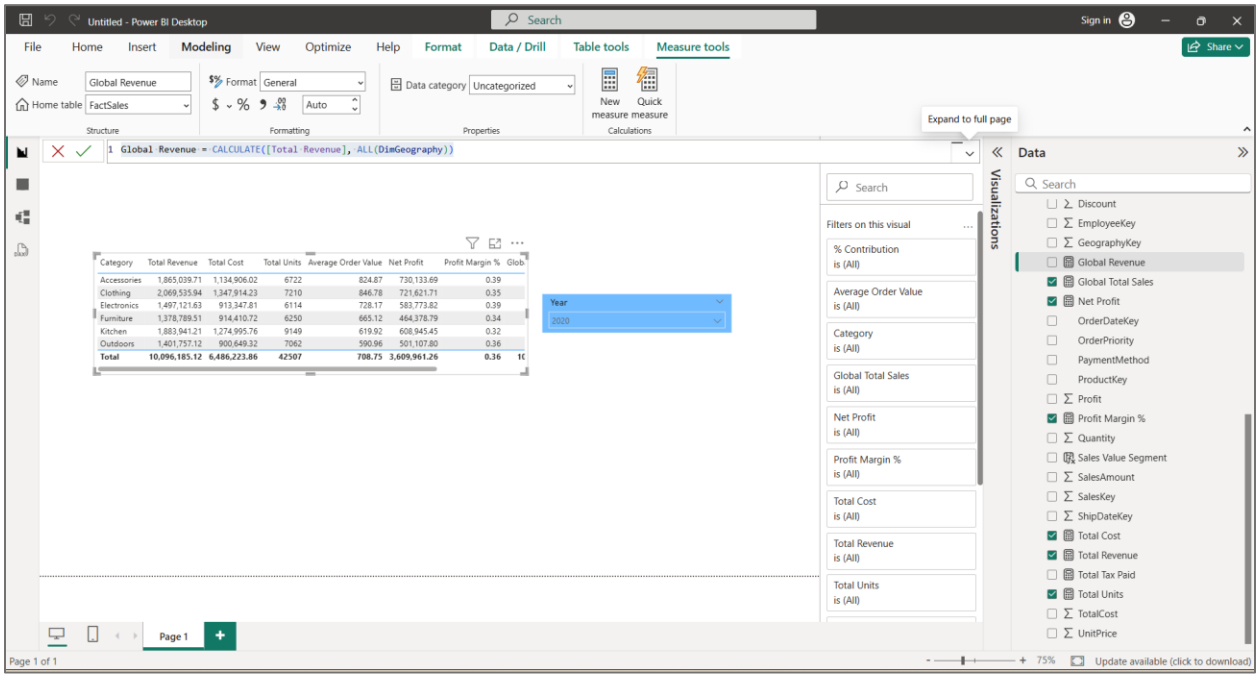


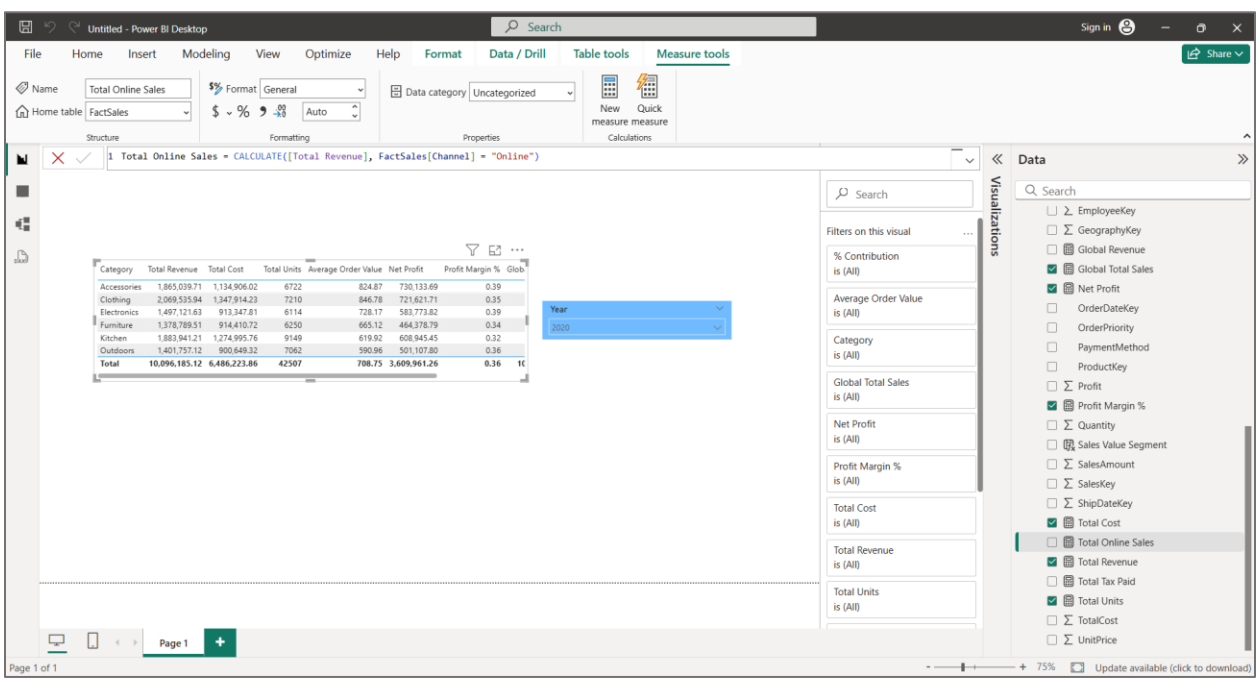


SECTION D: FILTER CONTEXT & CALCULATE

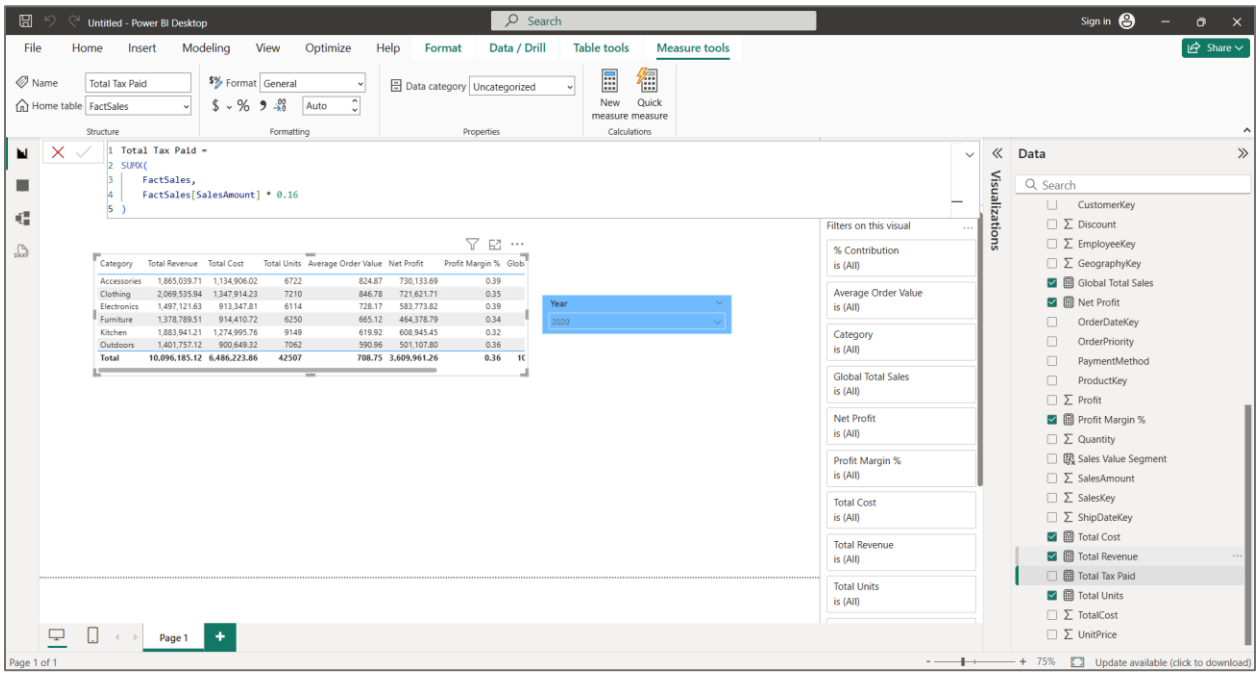
By using the CALCULATE function combined with ALL(), I successfully demonstrated Context Override. While the standard Total Revenue measure respects the 'Filter Context' (responding to row labels and slicers), the Global Revenue measure intentionally ignores these filters. This allows

for a direct comparison between a specific segment and the entire business, which is essential for calculating the % Contribution metric.

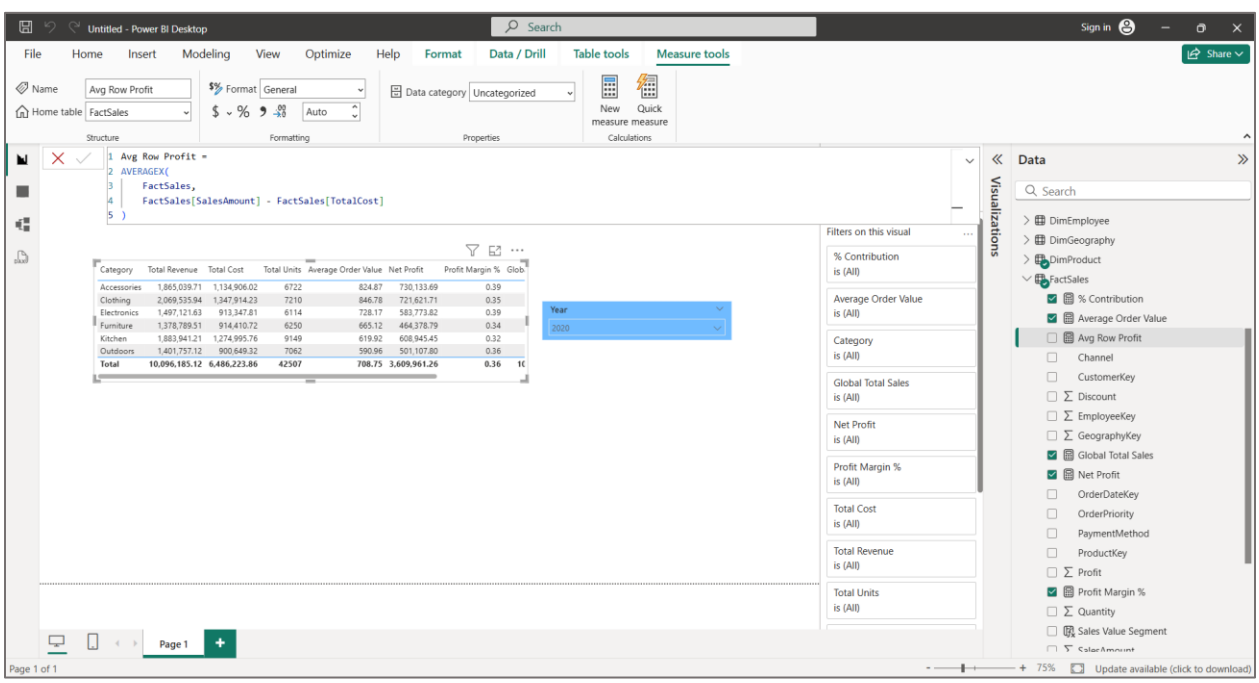




SECTION E: ITERATOR FUNCTION

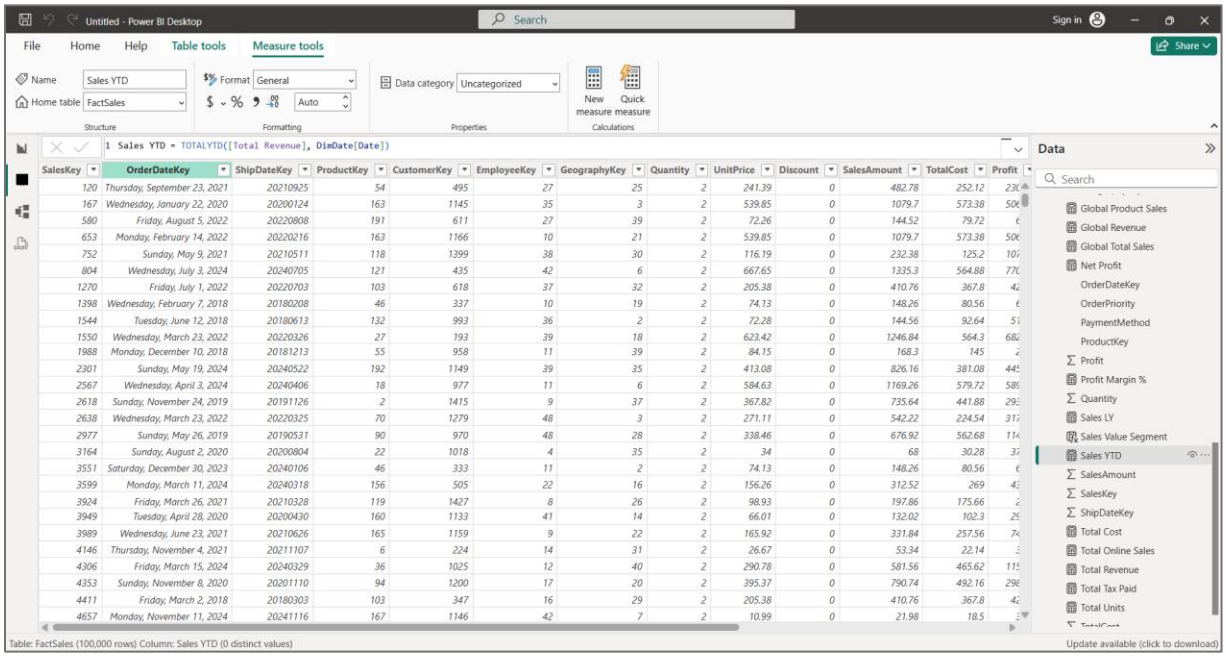


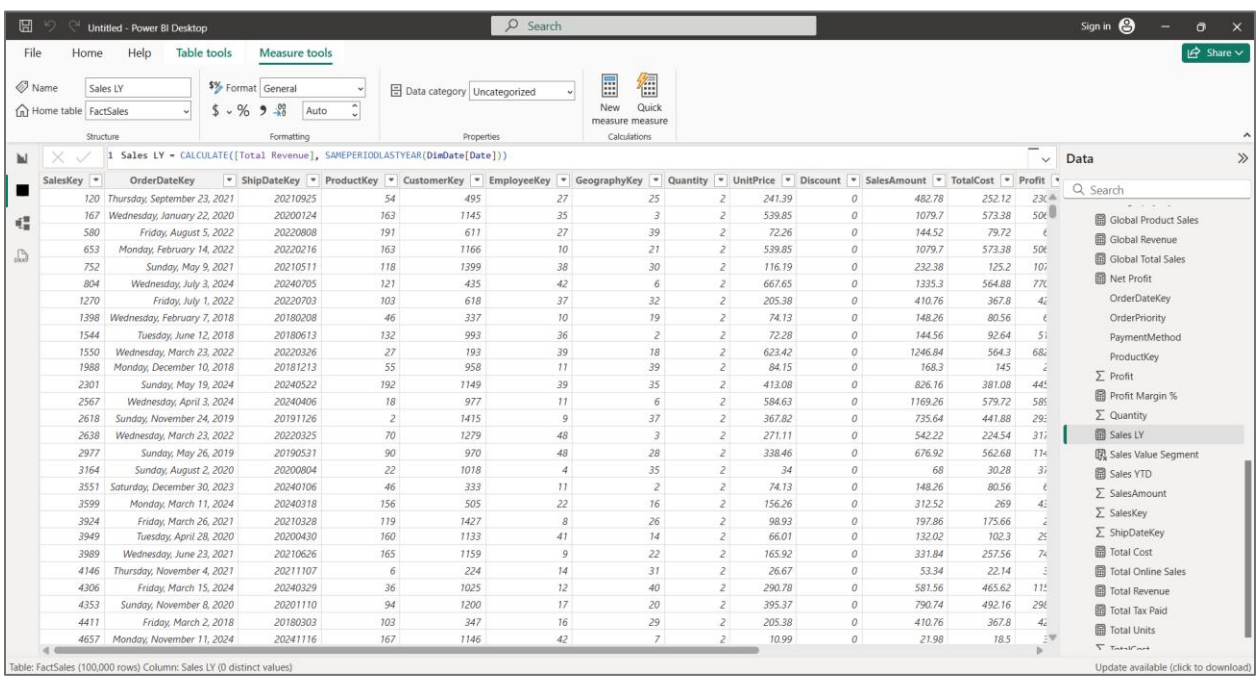
The Total Tax Paid measure utilizes SUMX, an iterator function. Unlike standard aggregation functions that operate over a filter context, SUMX creates a Row Context. It iterates through the FactSales table line-by-line, calculating the 16% tax for each individual transaction before summing the results. This demonstrates an understanding of how row-level granularity can be maintained within a dynamic measure.

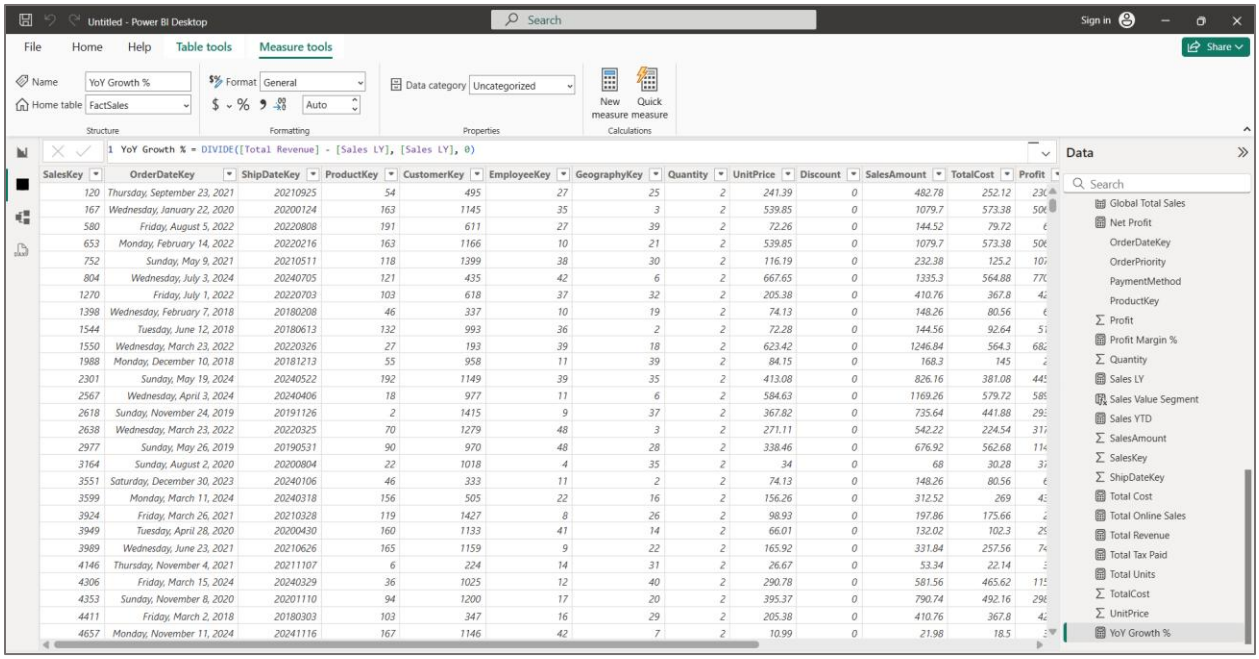


I implemented AVERAGEX to calculate the average profit per transaction. This function was chosen because 'Profit' is not a static column in the raw data. By using an iterator, I successfully invoked a **Row Context** that performs a row-level subtraction for every record in the FactSales table before aggregating the results into a final average. This demonstrates that the measure is sensitive to individual transaction granularity even when displayed at a high-level category view.

SECTION F: TIME INTELLIGENCE







I utilized the dedicated DimDate table to perform Time Intelligence calculations. By using TOTALYTD and SAMEPERIODLASTYEAR referenced against the DimDate[Date] column, I ensured that the calculations remain accurate across different calendar granularities and respond dynamically to the date slicer.

Year	Total Revenue	Sales LY	YoY Growth %
2023	1,627,100.12	1,505,034.44	0.08
Total	**1,627,100.12**	**1,505,034.44**	**0.08**

Year

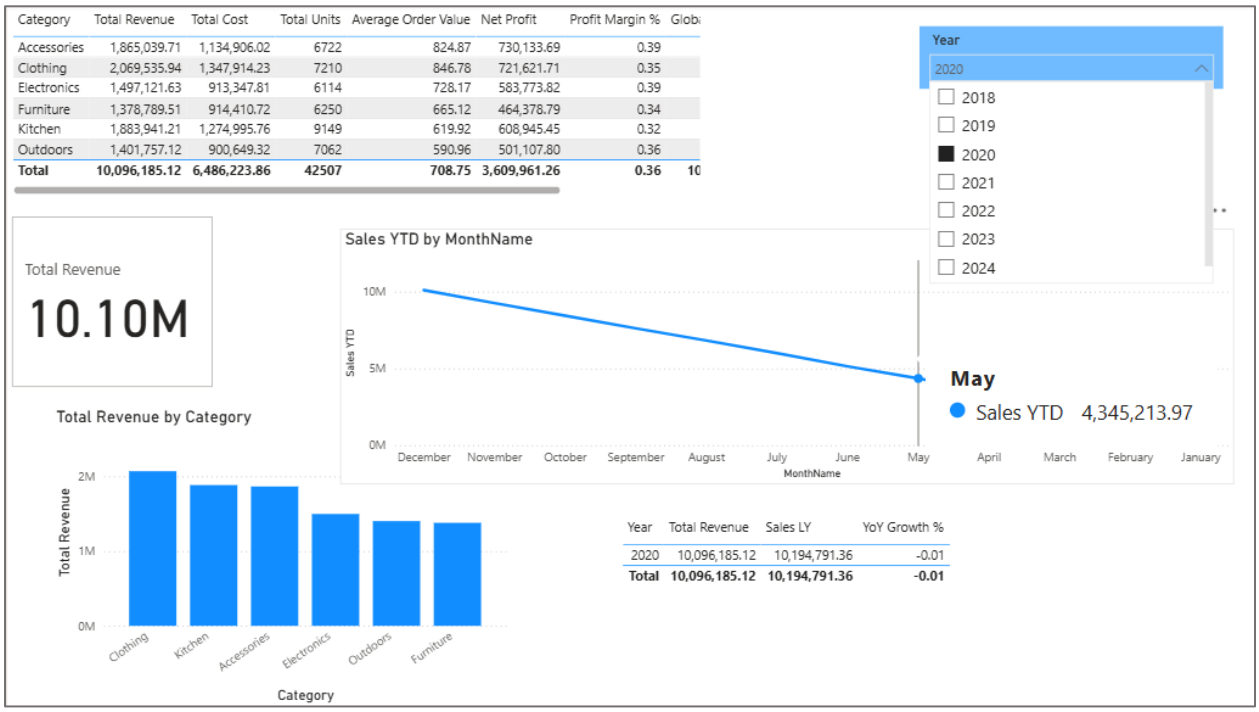
2023

Year	Total Revenue	Sales LY	YoY Growth %
2024	1,563,799.83	1,627,100.12	-0.04
Total	**1,563,799.83**	**1,627,100.12**	**-0.04**

Year

2024

SECTION G: VISUAL VALIDATION



Category	Total Revenue	Total Cost	Total Units	Average Order Value	Net Profit	Profit Margin %	Glob
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Clothing	2,069,535.94	1,347,914.23	7210	846.78	721,621.71	0.35	
Electronics	1,497,121.63	913,347.81	6114	728.17	583,773.82	0.39	
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Total	**10,096,185.12**	**6,486,223.86**	**42507**	**708.75**	**3,609,961.26**	**0.36**	**10**

Year	Total Revenue	Sales LY	YoY Growth %
2020	10,096,185.12	10,194,791.36	-0.01
Total	**10,096,185.12**	**10,194,791.36**	**-0.01**

GitHub Link

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